

The Junior exam consists of problems **1 - 30**. The Senior exam consists of problems **11 - 40**. Students had **75 minutes** to complete the exam.

Problem 1.

Which prime number, when you take the sum of it and its two neighbours, gives a perfect square?

Answer: 3

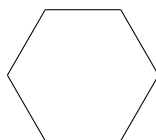
Problem 2.

The SMO has 3 large and 3 small plushies, all of which are different. They want to create a mascot family with 1 large and 2 small plushies. In how many ways can they create their mascot family?

Answer: 9

Problem 3.

Given a regular hexagon, how many different triangles can you create using its vertices? (Two triangles are different if one has a vertex that the other doesn't have).



Answer: 20

Problem 4.

Enter each of the numbers $1, \dots, 5$ into one of the boxes below so that the result is as large as possible.

$$\frac{\square \cdot (\square + \square)}{\square + 2 \cdot \square}$$

Now write the result as a reduced fraction (if it's already in lowest terms, leave it as it is). What is the sum of the resulting numerator and denominator?

Answer: 39

Problem 5.

Felix wants to create numbers from the names of his friends. In each name, different letters stand for different digits, and identical letters for the same digit. Which name can he turn into the biggest number?

- a) Sarah b) Anna c) Marco d) Belle e) David

Answer: Sarah

Problem 6.

Irina plays a game: each day she can visit one of the islands A , B or C , and she is not allowed to visit the same island two days in a row. At island A she gets a coconut every time she visits, at island B she gets a coconut every second time she visits and at island C she gets two coconuts every third time she visits. If Irina wants to collect a total of 14 coconuts as quickly as possible, how many days would she need?

- a) 8 b) 14 c) 17 d) 20 e) 24

Answer: 17

Problem 7.

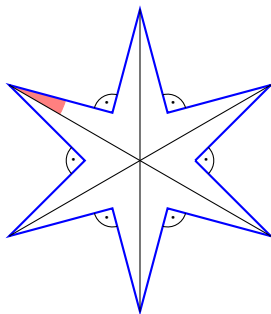
Mark is running laps on a circular track with a radius of 50 metres. He has placed water stations along the track. As he runs along the track, he realizes that after each station, he has to run for at least 105 metres in order to reach the next station. What is the maximum number of water stations Mark could have placed?

- a) 2 b) 3 c) 4 d) 5 e) 6

Answer: 2

Problem 8.

In the given picture, the tips of the star form a regular hexagon, and all blue segments have the same length. The marked angles are 90 degrees.



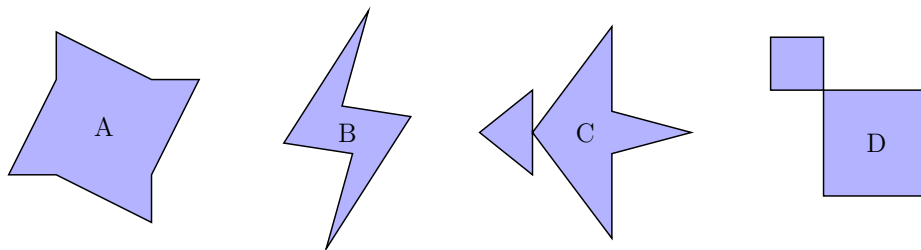
How many degrees is the red angle?

- a) 12 b) 15 c) 20 d) 24 e) 30

Answer: 15

Problem 9.

Which of the following shapes have an axis of reflective symmetry?



Answer: C, D

Problem 10.

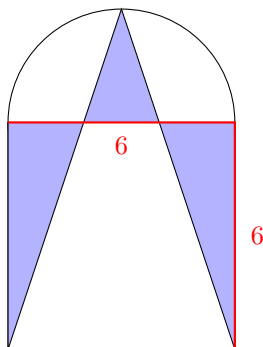
We partition the numbers $1, \dots, 10$ into five pairs. Which of the following statements could be true?

- A The numbers in every pair have the same sum.
- B The numbers in every pair have a difference of 2.
- C Every pair has at least one prime number.
- D Every pair has at least one odd number.

Answer: A, D

Problem 11.

What is the area of the shaded region? The entire shape is a semicircle on top of a square with side length 6.



Answer: 15

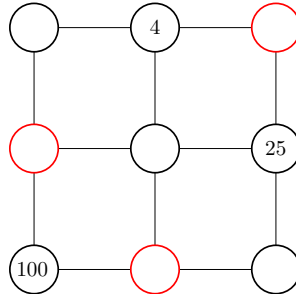
Problem 12.

Eleven people stand along a straight path. Simultaneously, they start walking along the path, all with the same speed but not necessarily in the same direction. Whenever two people cross, they do a high-five. What is the largest possible number of high-fives?

Answer: 30

Problem 13.

A positive integer is a whole number greater than zero: $1, 2, 3, \dots$ and so forth. In the grid below, Evelyn wants to write distinct positive integers inside each circle, such that for each of the four small squares, the numbers inside the corners multiply to 1000. She already wrote down three numbers. What is the sum of the numbers in the red circles?



Answer: 17

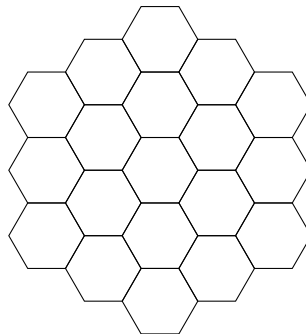
Problem 14.

Artur has placed a regular hexagon on the blackboard and traced its outline with chalk. Afterwards, every second, he rotates the hexagon clockwise around its center by 40° . After how many seconds does the hexagon first match the drawn outline again?

Answer: 3

Problem 15.

In the beehive depicted below, some hexagonal cells should be filled with honey. At least how many cells need to be filled such that each empty cell touches at least one cell containing honey?



a) 3

b) 4

c) 5

d) 6

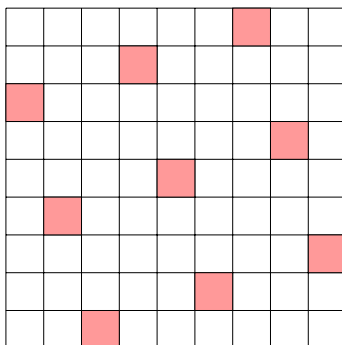
e) 7

Answer: 4

Problem 16.

Nine squares on a 9×9 board are coloured red, as shown below. The uncoloured squares should be covered by rectangles (possibly with different dimensions) such that each uncoloured square is covered by exactly one rectangle, but none of the rectangles covers a red square.

At least how many rectangles are needed?

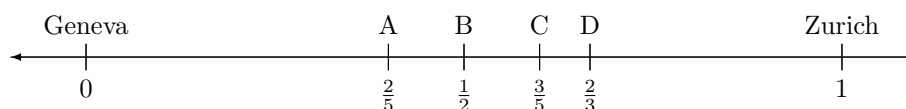


- a) 12 b) 13 c) 14 d) 15 e) 16

Answer: 12

Problem 17.

The Swiss Mathematical Olympiad is organizing an event somewhere along the segment between Zurich and Geneva for 15 people from Zurich and 10 people from Geneva. Where should they organize the event such that the total distance traveled by everyone is minimized?

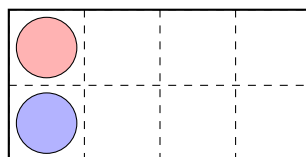


- a) A b) B c) C d) D e) Zurich

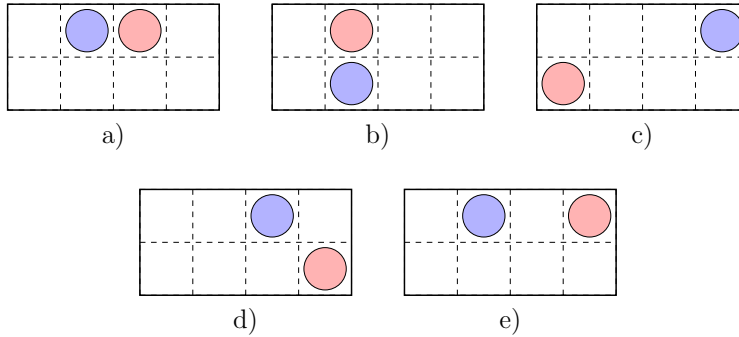
Answer: In Zurich

Problem 18.

Two robots are placed on a 2×4 board, as shown below. In a turn, either robot can move in one of the four cardinal directions and only stops once it hits a wall or the other robot (it will always stop on a free unit square).



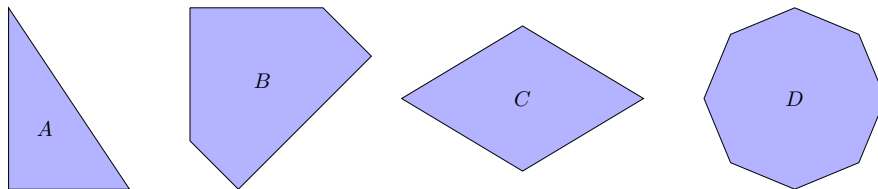
Starting from the configuration above, which of the following configurations cannot be reached?



Answer: b)

Problem 19.

Which of the following shapes can be formed by the intersection of two squares?



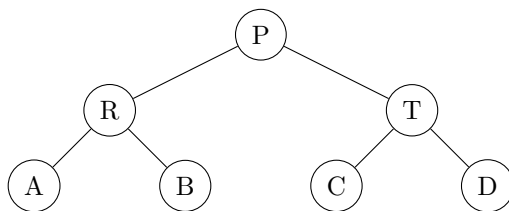
Answer: A, B, D

Problem 20.

Depicted below is a family tree consisting of grandfather Paul, his children Raphi and Tanish and his grandchildren, Adil, Barbara, Clemens and David. Each of them is either a truth-teller or a liar. Here is what they say:

- Paul: Exactly one of my children is a liar. Both my children would say the same about their own children.
- Adil: My father is a liar!
- Barbara: My brother is liar!
- Clemens: Among us seven, an even number are liars.
- David: Exactly one of my cousins is lying.

Knowing that Paul is a truth-teller, which of the following is always true ?



- A Adil is lying.
- B Barbara is lying.
- C Clemens is lying.
- D David is lying.

Answer: A

Problem 21.

The numbers $1, 2, 3, \dots, 49$ are written on a 7×7 grid. We have the following.

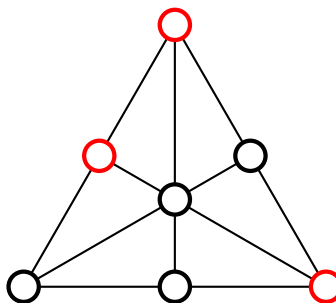
1. The grid is coloured with a chessboard colouring (i.e. every square is black or white; white squares only share edges with black squares and vice versa).
2. The four corner squares are all white.
3. The number 1 is on a white square.
4. Consecutive numbers (e.g. 25, 26) always appear in two squares that share an edge.

What is the sum of the white squares minus the sum of the black squares?

Answer: 25

Problem 22.

In the following triangle, there are 7 unknown numbers in the circles. But Tom knows that the sum of the numbers on any line is 17. What is the sum of the numbers in the red circles?



Answer: 17

Problem 23.

Multiplying the digits of his favorite positive integer 321738, Samuel obtains the result 1008. The digits of Raphi's favourite positive integer also multiply to 1008. What is the smallest positive integer that could be Raphi's favourite?

Answer: 2789

Problem 24.

How many ways are there to choose two integers $1 \leq a < b \leq 20$ such that b is a multiple of a ?

Answer: $10+6+5+4+3+2+2+2+2+10 = 46 = 19+9+5+4+3+2+1+1+1+1$

Problem 25.

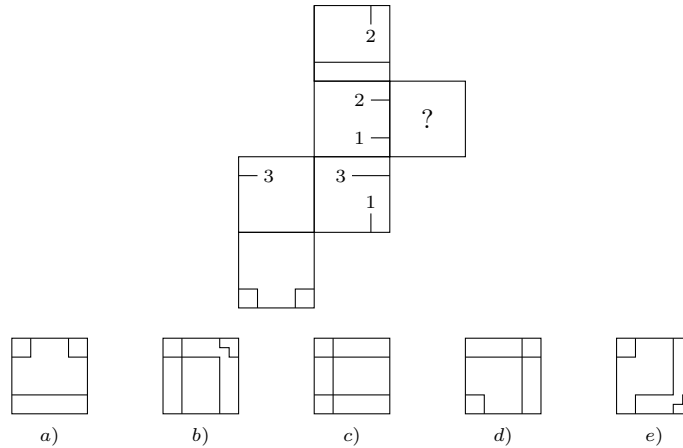
Leonhard has a secret treasure chest with a dual lock. If 2 good keys are inserted, the chest opens, otherwise nothing happens. There are 7 keys in total, 4 of which are good. Mark steals the keys; however, he does not know which ones are good. What is the least number of attempts Mark needs to guarantee that he can open the chest?

- a) 4 b) 5 c) 6 d) 7 e) 8

Answer: 5

Problem 26.

The following net is folded into a cube. Which pattern (possibly rotated) has to go into the square with the “?” so that the pairs of equal numbers are connected?



Answer: c)

Problem 27.

Adil wants to place either a multiplication sign ($\times$) or a division sign ($\div$) in each of the boxes below, and to compute the resulting expression:

$$1 \square 2 \square 3 \square 4 \square 5 \square 6 \square 7 \square 8 \square 9 \square 10$$

What is the smallest integer he can get?

- a) 1 b) 7 c) 10 d) 14 e) 28

Answer: 7

Problem 28.

At most how many bishops can you place on a standard 8×8 chessboard such that no two bishops are on the same diagonal? (A diagonal is any set of squares whose centres lie on a line that makes an angle of 45° with the sides of the board.)

- a) 12 b) 13 c) 14 d) 15 e) 16

Answer: 14

Problem 29.

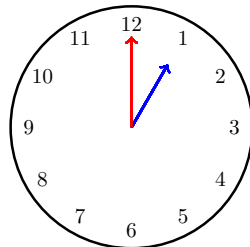
Let m be a positive integer. Tanish has arranged m rectangles, each of dimensions 1×2 , to form a solid $n \times n$ square. Which of the following must be true?

- A m is even.
 B n is even.
 C $m \neq 2026$.
 D $m > n$.

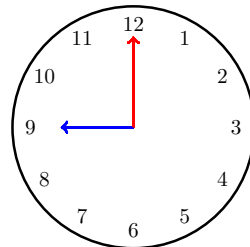
Answer: A, B, C

Problem 30.

The hour hand (blue) of Florian's broken clock is moving too slowly but the minute hand (red) is fine. Both move at a constant speed. Florian observed the clock at two different times in the same week:



Monday, 10:00



Tuesday, 16:00

Which number(s) could the hour hand point to on Wednesday, 7:00 that week?

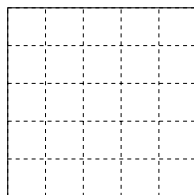
- A 1
- B 4
- C 7
- D 10

Answer: A, C

Problem 31.

On a square piece of paper, a 5×5 grid is drawn. Emily cuts a rectangle out of the piece of paper. She cuts only along the grid lines and notes that the total perimeter (including holes) of the remaining shape is the same as the perimeter of the original square.

In how many ways could she have cut out the rectangle?



Answer: 64

Problem 32.

Marco wants to travel from the bottom-left to the top-right unit square of a 2×10 board. In a step, he can move between squares that share an edge, and he is not allowed to visit a square more than once. How many different paths can Marco take?

Answer: 512

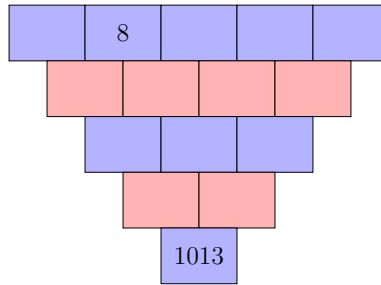
Problem 33.

Francesc has 10 identical white chairs, 10 identical red chairs, and 1 blue chair. He chooses 12 chairs and places them around a table such that no two chairs of the same colour have exactly one chair between them. In how many ways could Francesc place the chairs around the table? (Two configurations are considered the same if and only if one can be turned into the other by rotating, but not reflecting.)

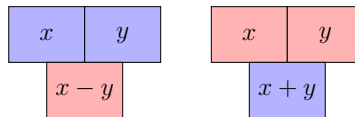
Answer: 5

Problem 34.

Ricardo draws an inverted pyramid coloured in blue and red as follows.



He starts by putting the numbers 8 and 1013 in his pyramid, as depicted in the sketch. He then fills the other squares with numbers such that the pyramid fulfils the following rules:



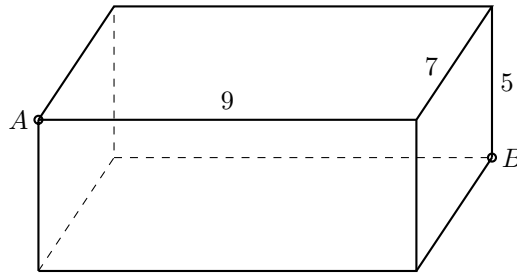
Elia comes along and replaces the 8 in the top row by a 16. He wants to keep the rest of the numbers in the top row the same, and change the others so that the rules are followed again.

What will he then end up putting in the bottommost square?

Answer: 1013

Problem 35.

Turbo the snail is crawling on the surface of the box depicted below. He starts at vertex A and wants to go to vertex B . How long is the shortest possible path that Turbo can take?

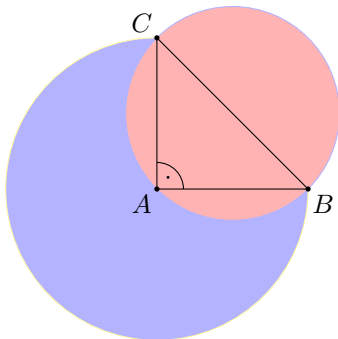


- a) 21 b) $\sqrt{130} + 5$ c) 17 d) $7\sqrt{5}$ e) 15

Answer: 15

Problem 36.

In the following diagram, A is the center of the blue circle and B, C are points lying on the edge of the circle such that $\angle BAC = 90^\circ$. A, B, C all lie on the edge of the red circle.



Given that $\overline{AB} = \overline{AC} = 4$, what is the area of the blue shape?

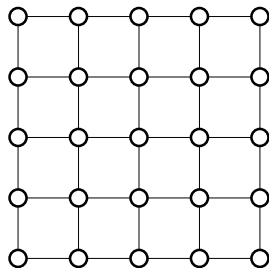
- a) 12π b) 8π c) $12\pi - 2\sqrt{2}$ d) $10\pi + \sqrt{2}$ e) $8\pi + 8$

Answer: $8\pi + 8$

Problem 37.

In the grid below, Lena colours some of the circles red and the others blue. She then erases all segments between red and blue circles. After that, she notices that between any two blue circles, there is a unique way to get from one to the other by following edges.

What is the maximum number of nodes that Lena coloured blue?

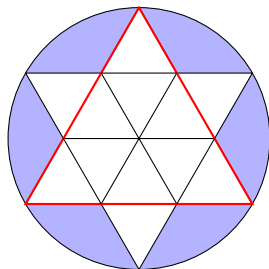


- a) 16 b) 17 c) 18 d) 19 e) 20

Answer: 19

Problem 38.

Depicted below is a shape made of a circle and equilateral triangles. The red triangle has side length 15. What is the total blue area?



- a) 50π b) $75(\pi - \sqrt{3})$ c) 75π d) $75\sqrt{3}$ e) $50(2\pi - 3)$

Answer: $75(\pi - \sqrt{3})$

Problem 39.

Veronica has written 5 different prime numbers on the blackboard. Which statements are always true?

- A The sum of the 5 numbers is odd.
- B There are 3 numbers whose sum is divisible by 3.
- C One of the numbers is larger than 10.
- D There are two numbers whose difference is divisible by 6.

Answer: B, C, D

Problem 40.

There is some positive integer $n > 1$ written on the blackboard. Noelia plays the following game: If n is not a prime number, she computes the two largest proper divisors of n (not including n itself) and replaces the number on the board by their sum. If there is a prime number on the board, the game stops. Which of the following statements are always true?

- A If in a step, the number n is replaced by the number m , then $m > \sqrt{n}$ must hold.
- B There exists a number n such that the game goes on indefinitely.
- C If n is 6^k , where $k \geq 2$ is an integer, then at some point 6 will be on the board.
- D If n is divisible by 2, 3 and 5, but no other primes, then at some point 6 will be on the board.

Answer: A, C